# The second law

**antocciphysics**



**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Explain the direction of heat transfer and the limits of energy conversion.

**SUCCESS CRITERIA - I can...**

1. Predict spontaneous net heat flow.
2. Explain why moving heat from cold to hot needs an additional process.
3. Distinguish energy conservation from process direction.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

## 01 Warm-up: bring back an earlier idea

Samples at 13°C and 55°C are mixed with no energy input or phase change. A student reports 67°C at equilibrium. Is that plausible? Explain.

**Marzano check - circle:** ① ② ③

## 02 Worked example: explain the model

**Reference:** $Q_{hot} = Q_{cold} + W_{in}$.
Refrigerator cycle: heat amounts are positive magnitudes. Work input moves heat from cold to hot; spontaneous flow is hot to cold.

**Worked reasoning:** Hot and cold blocks in contact exchange energy until thermal equilibrium. Total energy can be conserved while the temperature difference and ability to drive a heat engine decrease.

**Your turn:** Draw before-and-after temperature states and the spontaneous heat-flow direction.

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

**Marzano check - circle:** ① ② ③

## 03 Read and interpret the model

A 50°C block touches a 10°C block. Draw net heat flow and predict their trend toward equilibrium.



50°C 10°C

Each outline represents the labeled sample or container.

**Marzano check - circle:** ① ② ③

## 04 Apply the physics

During one modeled refrigerator cycle, 20 J is removed from the cold compartment and 5 J of electrical work enters. Draw and label the energy flows. How much energy must be released to the room?

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

**Marzano check - circle:** ① ② ③

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Continue down each column.

## 05 Explain your reasoning

A proposed machine takes heat from one reservoir and turns all of it into work in a repeating cycle. What thermodynamic limitation is missing?

**Marzano check - circle:** 1 2 3

## 06 Review: connect an earlier idea

An energy account begins with 24 J. It ends with 16 J motion energy and 4 J thermal energy. Find the unaccounted energy and suggest a next check. Show the energy changes and transfers; label units.

| Initial | Transfers | Final |
|---|---|---|
| | | |

Draw bars or transfer arrows; label forms and joules.

**Marzano check - circle:** 1 2 3

## 07 Exit check

Does a refrigerator disprove hot-to-cold spontaneous flow? Explain the role of work input.

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

**Marzano check - circle:** 1 2 3

## 08 STRETCH: test the model

A passive cooling pack and a powered refrigerator both cool objects. Draw the energy transfers for each. Explain why moving energy from a colder compartment to a warmer room requires an input in the refrigerator case.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

**Marzano check - circle:** 1 2 3